

HW5_r

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Problem 4

```
library(ggplot2)
```

```
spotify <- read.csv('ds4420_spotify.csv')
head(spotify)
```

```
##           song_title      artist_name release_month
## 1      double take          Dhruv              1
## 2    One Last Kiss    Hikaru Utada              3
## 3         Autumn      NIKI                  8
## 4 Sweet Heat Lightning Gregory Alan Isakov      8
## 5 Brothers In Arms - Remastered 1996    Dire Straits      5
## 6          Fuego      Pitbull             10
##  release_year artist_pop track_pop duration_s explicit
## 1         2022         66        78    171.743    FALSE
## 2         2021         73        63    252.026    FALSE
## 3         2022         79        72    232.800    FALSE
## 4         2023         73        78    286.120    FALSE
## 5         1985         81        74    420.240    FALSE
## 6         2006         88        57    229.213     TRUE
```

```
explicit <- spotify$explicit
n <- length(explicit)
y <- sum(explicit)
alpha <- 2
beta <- 18
```

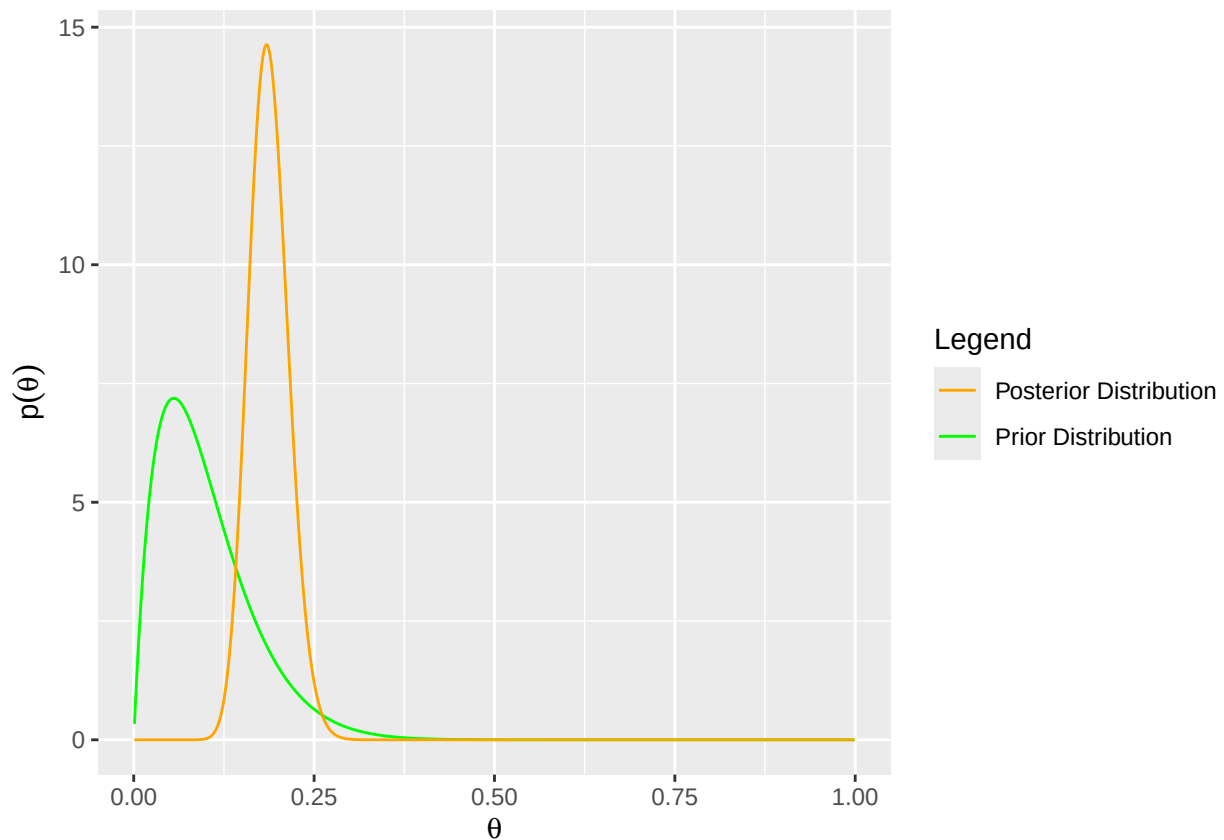
```
prior <- function(theta, alpha, beta) {
  log_num <- (alpha - 1) * log(theta) + (beta - 1) * log(1 - theta)
  log_den <- lbeta(alpha, beta)
  exp(log_num - log_den)
}

posterior <- function(theta, alpha, beta, y, n) {
  alpha_star <- alpha + y
  beta_star <- beta + n - y
  log_num <- (alpha_star - 1) * log(theta) + (beta_star - 1) * log(1 - theta)
  log_den <- lbeta(alpha_star, beta_star)
}
```

```
    exp(log_num - log_den)
  }
```

```
sim_theta <- seq(0.001, 0.999, length.out = 1000)
prior_dist <- prior(sim_theta, alpha, beta)
post_dist <- posterior(sim_theta, alpha, beta, y, n)

plot_df <- data.frame(theta = sim_theta, Prior = prior_dist, Posterior = post_dist)
ggplot(plot_df, aes(x = theta)) +
  geom_line(aes(y = Prior, color = 'Prior Distribution')) +
  geom_line(aes(y = Posterior, color = 'Posterior Distribution')) +
  labs(x = expression(theta), y = expression(p(theta))) +
  scale_color_manual(name = "Legend", values = c(
    "Prior Distribution" = "green",
    "Posterior Distribution" = "orange"
  ))
))
```



```
prior_mean <- alpha / (alpha + beta)
MLE <- y / n
posterior_mean <- (alpha + y) / (alpha + beta + n)

prior_mean
```

```
## [1] 0.1
```

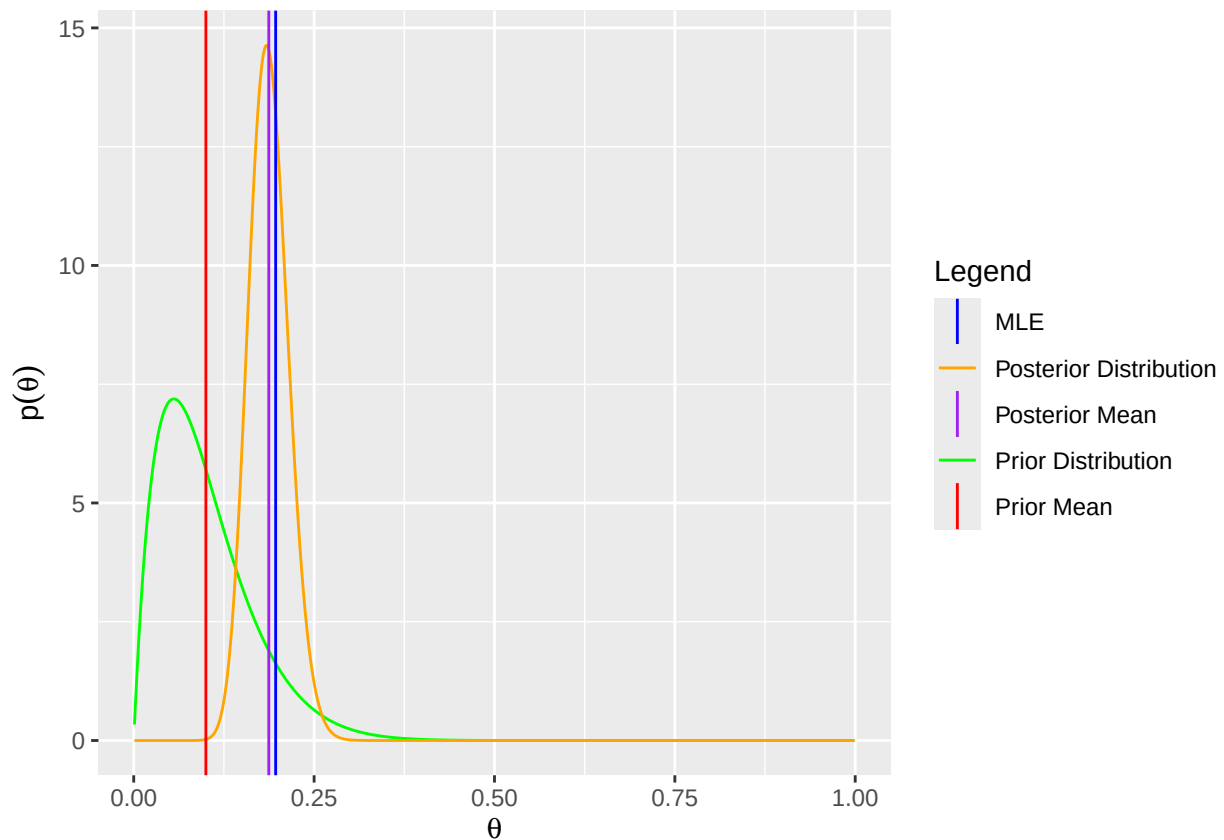
```
MLE
```

```
## [1] 0.1967213
```

```
posterior_mean
```

```
## [1] 0.1871921
```

```
ggplot(plot_df, aes(x = theta)) +  
  geom_line(aes(y = Prior, color = "Prior Distribution")) +  
  geom_line(aes(y = Posterior, color = "Posterior Distribution")) +  
  geom_vline(aes(xintercept = prior_mean, color = "Prior Mean")) +  
  geom_vline(aes(xintercept = MLE, color = "MLE")) +  
  geom_vline(aes(xintercept = posterior_mean, color = "Posterior Mean")) +  
  labs(x = expression(theta), y = expression(p(theta))) +  
  scale_color_manual(name = "Legend", values = c(  
    "Prior Distribution" = "green",  
    "Posterior Distribution" = "orange",  
    "Prior Mean" = "red",  
    "MLE" = "blue",  
    "Posterior Mean" = "purple"  
  ))
```



The posterior mean being in between the prior and MLE makes sense as it is a weighted average of the prior and MLE. As n , the number of observations in our data set, grows, our posterior will converge to our MLE.

The prior mean is just our initial belief before using the data, while the MLE represents the data. We would use our posterior mean as our estimate for theta. So the probability of a song being explicit would be our posterior mean. To validate the performance of our model, we can train test split the data, use the train data to create a posterior, and compare the posterior mean to the percentage of explicit music in our test data set.

Problem 5 Part

```
track_pop <- spotify$track_pop
n <- length(track_pop)
y_bar <- mean(track_pop)
sigma <- 20
eta <- 50
tao <- 2
mu_star <- (((n * y_bar) / sigma^2) + (eta / tao^2)) / ((n / sigma^2) + (1 / tao^2))
sigma_star <- sqrt(((n / sigma^2) + (1 / tao^2))^-1)

prior <- function(mu, eta, tao) {
  (1 / (tao * sqrt(2 * pi))) * exp(-(mu - eta)^2 / (2 * tao^2))
}

posterior <- function(mu, mu_star, sigma_star) {
  prior(mu, mu_star, sigma_star)
}

prior_mean <- eta
MLE <- y_bar
posterior_mean <- mu_star

prior_mean
```

```
## [1] 50
```

```
MLE
```

```
## [1] 65.99454
```

```
posterior_mean
```

```
## [1] 60.34276
```

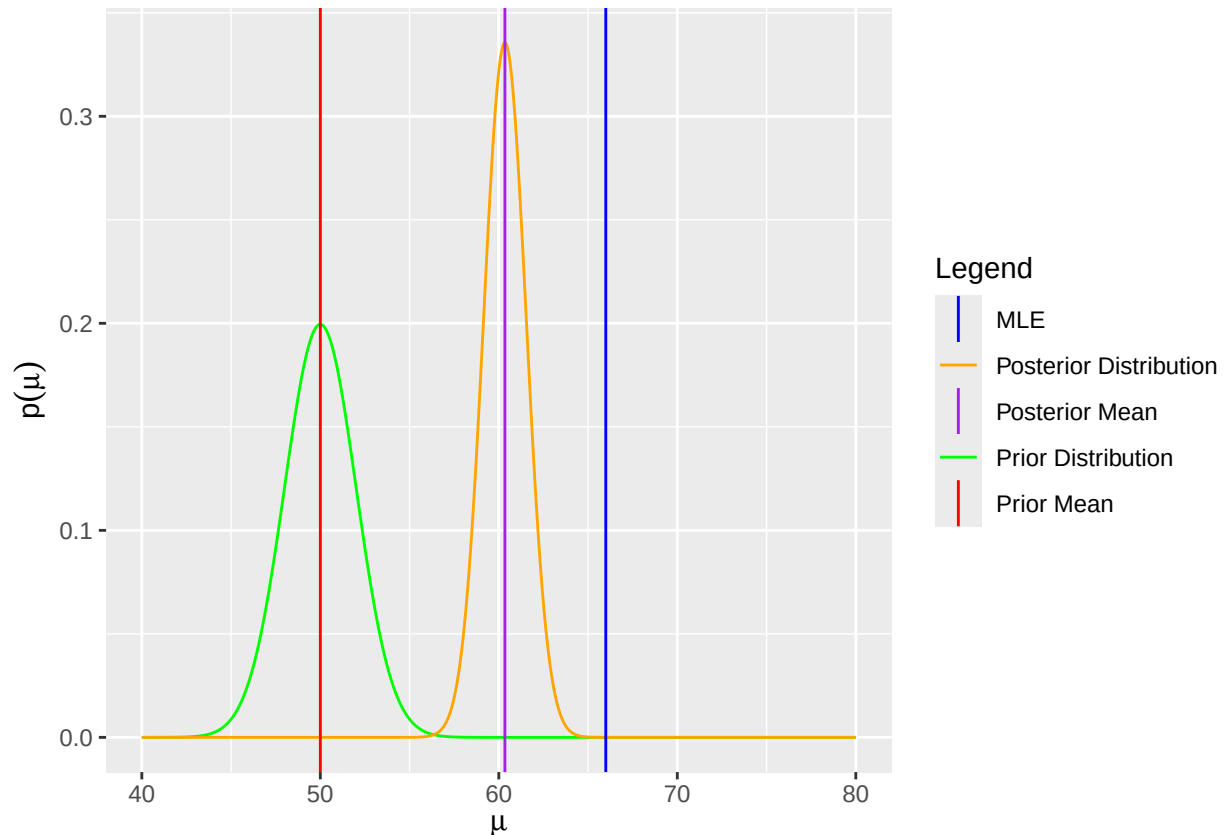
```
sim_mu <- seq(40, 80, length.out = 1000)
prior_dist <- prior(sim_mu, eta, tao)
post_dist <- posterior(sim_mu, mu_star, sigma_star)

plot_df <- data.frame(mu = sim_mu, Prior = prior_dist, Posterior = post_dist)
ggplot(plot_df, aes(x = mu)) +
  geom_line(aes(y = Prior, color = "Prior Distribution")) +
  geom_line(aes(y = Posterior, color = "Posterior Distribution")) +
  geom_vline(aes(xintercept = prior_mean, color = "Prior Mean")) +
```

```

geom_vline(aes(xintercept = MLE, color = "MLE")) +
geom_vline(aes(xintercept = posterior_mean, color = "Posterior Mean")) +
labs(x = expression(mu), y = expression(p(mu))) +
scale_color_manual(name = "Legend", values = c(
  "Prior Distribution" = "green",
  "Posterior Distribution" = "orange",
  "Prior Mean" = "red",
  "MLE" = "blue",
  "Posterior Mean" = "purple"
))

```



The posterior mean is close to the MLE as the large dataset weighs the posterior to be closer to the MLE. The prior mean is our initial belief before data, the MLE reflects our data, and the posterior utilizes both our initial belief and our data.

```

eta <- 75
tao <- 2
mu_star <- (((n * y_bar) / sigma^2) + (eta / tao^2)) / ((n / sigma^2) + (1 / tao^2))
sigma_star <- sqrt(((n / sigma^2) + (1 / tao^2))^-1)

prior <- function(mu, eta, tao) {
  (1 / (tao * sqrt(2 * pi))) * exp(-(mu - eta)^2 / (2 * tao^2))
}

posterior <- function(mu, mu_star, sigma_star) {
  prior(mu, mu_star, sigma_star)
}

```

```

}

prior_mean <- eta
MLE <- y_bar
posterior_mean <- mu_star

prior_mean

```

```
## [1] 75
```

```
MLE
```

```
## [1] 65.99454
```

```
posterior_mean
```

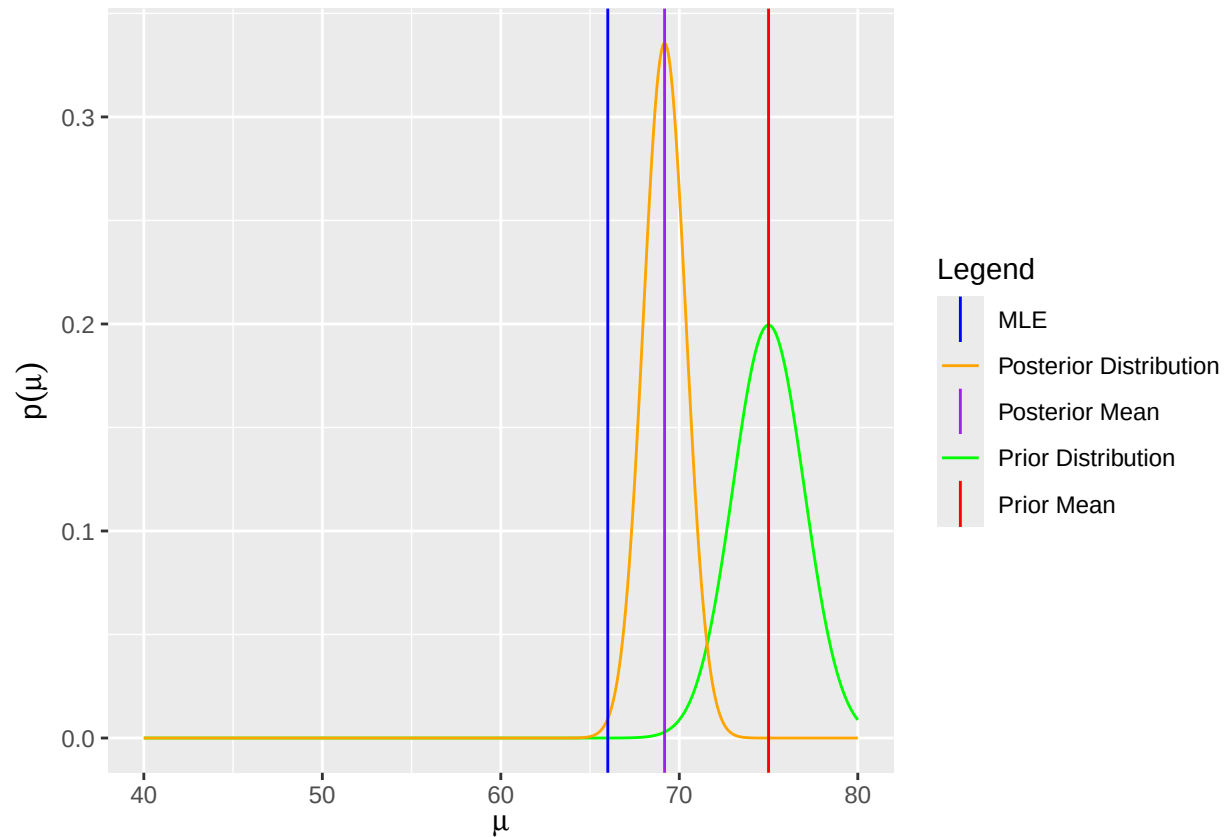
```
## [1] 69.17668
```

```

sim_mu <- seq(40, 80, length.out = 1000)
prior_dist <- prior(sim_mu, eta, tao)
post_dist <- posterior(sim_mu, mu_star, sigma_star)

plot_df <- data.frame(mu = sim_mu, Prior = prior_dist, Posterior = post_dist)
ggplot(plot_df, aes(x = mu)) +
  geom_line(aes(y = Prior, color = "Prior Distribution")) +
  geom_line(aes(y = Posterior, color = "Posterior Distribution")) +
  geom_vline(aes(xintercept = prior_mean, color = "Prior Mean")) +
  geom_vline(aes(xintercept = MLE, color = "MLE")) +
  geom_vline(aes(xintercept = posterior_mean, color = "Posterior Mean")) +
  labs(x = expression(mu), y = expression(p(mu))) +
  scale_color_manual(name = "Legend", values = c(
    "Prior Distribution" = "green",
    "Posterior Distribution" = "orange",
    "Prior Mean" = "red",
    "MLE" = "blue",
    "Posterior Mean" = "purple"
  ))

```



The MLE is now on the left side because our η value is larger than our MLE. The posterior mean still leans close to our MLE.

```
eta <- 50
tao <- 5
mu_star <- (((n * y_bar) / sigma^2) + (eta / tao^2)) / ((n / sigma^2) + (1 / tao^2))
sigma_star <- sqrt(((n / sigma^2) + (1 / tao^2))^-1)

prior <- function(mu, eta, tao) {
  (1 / (tao * sqrt(2 * pi))) * exp(-(mu - eta)^2 / (2 * tao^2))
}

posterior <- function(mu, mu_star, sigma_star) {
  prior(mu, mu_star, sigma_star)
}

prior_mean <- eta
MLE <- y_bar
posterior_mean <- mu_star

prior_mean
```

```
## [1] 50
```

MLE

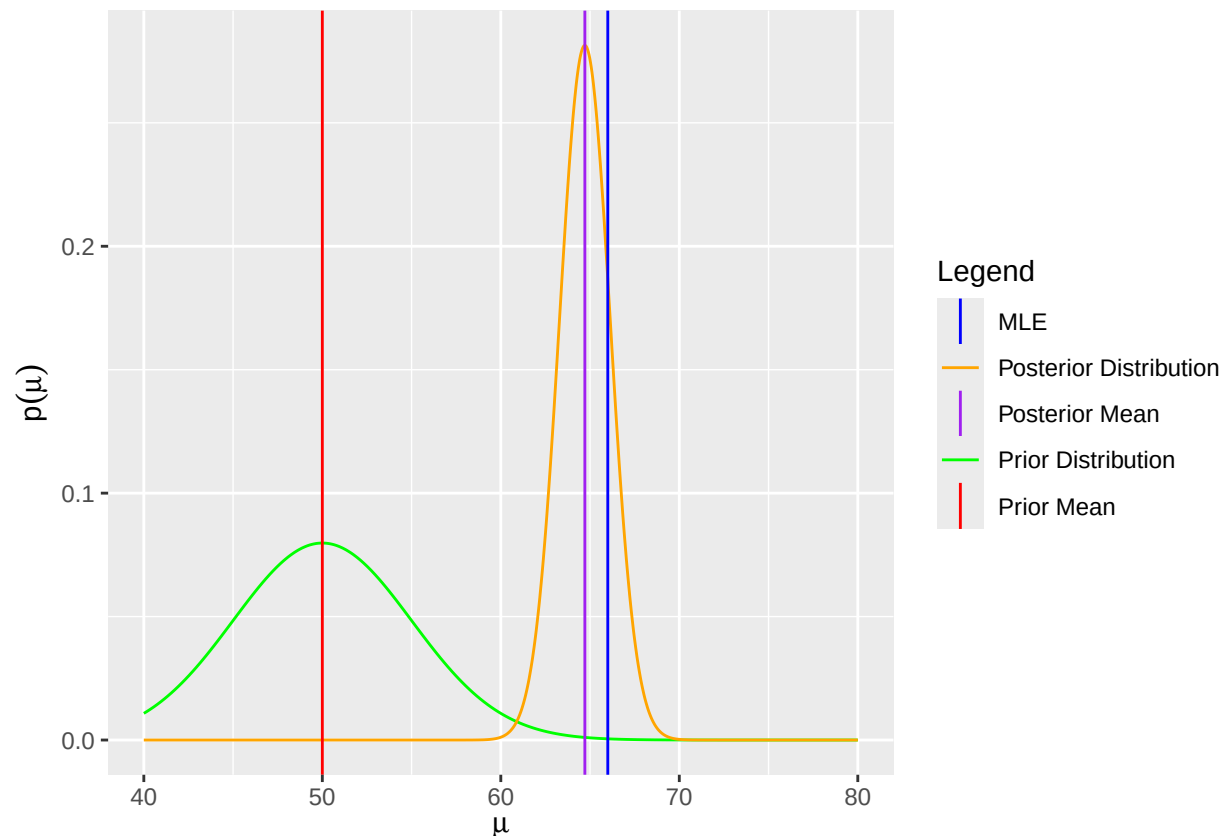
```
## [1] 65.99454
```

posterior_mean

```
## [1] 64.70854
```

```
sim_mu <- seq(40, 80, length.out = 1000)
prior_dist <- prior(sim_mu, eta, tao)
post_dist <- posterior(sim_mu, mu_star, sigma_star)

plot_df <- data.frame(mu = sim_mu, Prior = prior_dist, Posterior = post_dist)
ggplot(plot_df, aes(x = mu)) +
  geom_line(aes(y = Prior, color = "Prior Distribution")) +
  geom_line(aes(y = Posterior, color = "Posterior Distribution")) +
  geom_vline(aes(xintercept = prior_mean, color = "Prior Mean")) +
  geom_vline(aes(xintercept = MLE, color = "MLE")) +
  geom_vline(aes(xintercept = posterior_mean, color = "Posterior Mean")) +
  labs(x = expression(mu), y = expression(p(mu))) +
  scale_color_manual(name = "Legend", values = c(
    "Prior Distribution" = "green",
    "Posterior Distribution" = "orange",
    "Prior Mean" = "red",
    "MLE" = "blue",
    "Posterior Mean" = "purple"
  ))
```

Compared to the other two, our posterior mean is closest to our MLE. This is due to a wider prior distribution, allowing for the posterior distribution to be pulled towards our MLE much more.

Part_c

The prior distribution with $\eta = 75$ and $\tau = 2$ resulted in the more significantly different posterior. This was due to the prior mean being shifted so far from the MLE. This exercise helped me understand how shifting parameters results in different posteriors. Not necessarily better or worse, but different. By shifting the parameters, different posteriors result. Some more or less influenced by the prior. Some lean the posterior in different directions by shifting the prior mean. In practice, I would use a train test split to choose different hyperparameters and evaluate them to help me choose the best possible hyperparameters.